

Nathan Gardner Hattersley

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SUMMARY

Ph.D. candidate in Economics at the University of Texas at Austin with an M.S. in Statistics. Specialize in causal inference, pricing, demand estimation, and structural modeling using large-scale observational data. Experience spans production software engineering, antitrust litigation, policy research and machine learning.

EDUCATION

- University of Texas at Austin** Austin, TX
Ph.D. Economics (expected 2026); M.A. Economics Aug 2020–Present
- University of Arizona** Tucson, AZ
M.S. Statistics; B.A. Mathematics and Computer Science Aug 2013–May 2018

SKILLS

- Programming:** Python, R, Julia, SQL, TypeScript, Rust, Java, C, Stata, MATLAB, Bash
- Economics & Statistics:** Causal inference, structural modeling, demand estimation, machine learning (XGBoost), nonparametrics, numerical optimization
- Tools:** Pandas, tidyverse, Postgres/PostGIS, spatial analysis, Docker, Git, HPC, Make/Snakemake, L^AT_EX
- Languages:** English (native); Spanish (proficient)

PROFESSIONAL EXPERIENCE

- Texas Office of the Attorney General** Austin, TX
Economics Fellow Jul 2024–Jun 2025
 - Analyzed antitrust cases involving algorithmic collusion, online advertising auctions, and digital markets using industrial organization methods.
 - Drafted economic analyses and evaluated expert reports to support litigation strategy in complex antitrust cases.
 - Presented analyses to attorneys and translated complex quantitative methods into clear, actionable legal arguments.
- World Bank Group** Remote
Short-Term Consultant Feb 2021–Aug 2021
 - Developed statistical indices and quantitative metrics measuring political participation in Afghanistan, using large-scale survey data.
 - Developed algorithms for ordinal survey aggregation and PCA-based index construction.
 - Authored technical documentation and communication statistical methods and findings to economists, policy researchers, and other stakeholders.
 - Developed and led internal training on R, tidy data, functional programming, and reproducible workflows.
- JPMorgan Chase & Co.** Houston, TX
Software Engineer Jun 2018–Jun 2020
 - Developed and maintained production trading systems in Python and TypeScript supporting interest rate derivatives.
 - Designed regulatory workflows and trade models for new financial instruments.
 - Migrated legacy platform to microservice architecture using React, GraphQL, and WebSockets.
 - Collaborated with traders, product managers, and engineers, and quantitative researchers to prioritize requirements and deliver production software.

SELECTED RESEARCH

- Dealer Consolidation and Vertical Relationships:** Causal analysis of automobile dealership consolidation using data on millions of car sales and geospatial analysis to estimation effects of prices and product availability.
- Dynamic Pricing in Automotives:** Developed and estimated a dynamic structural model of manufacturer-dealer relationships. Combined structural estimation with machine learning (XGBoost) to recover underlying allocation and sales target policy of the manufacturer. Used estimates to recover welfare effects of dealer allocation policy.
- Demand Elasticity and Curvature (with Eugenio Miravete):** Nonparametric estimation of consumer demand to study the relationship between elasticity, curvature, pass-through, and market power. Contrasted nonparametric estimates with common parametric demand families and showed important implications for estimation and competition policy.

ADDITIONAL EXPERIENCE

- Teaching:** Teaching assistant for Master’s Econometrics, Industrial Organization, and Causal Inference. Led weekly review sessions, designed programming assignments, and taught statistical computing.
- Research Computing:** Led workshops on high-performance computing and reproducible research for Economics Ph.D. students. Developed departmental guidance for parallel computing in Julia, Python, and MATLAB.